A proof of Sun's ninety-dollar series for $1/\pi$

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Abstract

We prove the double-series identity for $90/\pi$ proposed by Zhi-Wei Sun in 2011. A restriction of its bivariate generating function satisfies a third-order differential equation and an additional first-order gradient identity. An algebraic substitution identifies the restricted period with a modular form of level 60. We prove the required general modular parametrization independently from eta products, cusp orders, and exact Sturm certificates of bounds 48 and 96. We also certify the degree-seven modular equation by Sturm's theorem, isolate the complex multiplication value by rational inequalities, and derive the weighted period evaluation directly by differentiating the Fricke transformation. No tabulated modular parametrization or special value is assumed. Exact rational-function and integer-series certificates accompany the proof.

1 The identity and the proof strategy

Sun's conjectural identity [1, Conjecture 5, (5.9)] is the following.

Theorem 1. *The series below converges absolutely, and*

$$\sum_{n=0}^{\infty}\frac{357n+103}{2160^n}\binom{2n}{n}\sum_{k=0}^{n}\binom{n}{k}\binom{n+2k}{2k}\binom{2k}{k}(-324)^{n-k}=\frac{90}{\pi}. \tag{1}$$

Sun offered ninety US dollars for the first rigorous proof. The identity was also listed as an open conjecture in the 2024 version of [2, Section 4, (4.3)]. This article supplies a proof and its computational certificates; it makes no claim that priority or a prize payment has been confirmed by the sponsor.

For any variable x, write $\vartheta_x = x\mathrm{d}/\mathrm{d}x$. The eta quotients and polynomials suggested by [3, Tables 2–4] provided candidate modular data; no identity from those tables is a theorem input here. Section 3 proves their required identities independently. The standard modular-form inputs are the eta-quotient modularity and cusp-order criteria [4, 5], the eta transformation under $\tau \mapsto -1/\tau$, and Sturm's theorem [6]. The needed derivative covariance is proved explicitly below. The CM value and weighted evaluation are also proved independently. The algebraic bridge consists of a restriction identity, a gradient relation, and an exact transformation of differential operators, each checked over a rational-function field.

2 A bivariate period and its restriction

Define, initially near $(u,v)=(0,0)$,

$$\mathcal{H}(u,v)=\sum_{j,k\ge 0}\binom{2j+2k}{j+k}\frac{(j+3k)!}{j!(k!)^3}u^j v^k. \tag{2}$$

Put $U=\vartheta_u\mathcal{H}$ and $V=\vartheta_v\mathcal{H}$. The substitution $j=n-k$ shows that the left side of (1) is

$$\bigl[357(U+V)+103\mathcal{H}\bigr]\left(-\frac{3}{20},\frac{1}{2160}\right). \tag{3}$$

Our restriction parameter is q, which is distinct from the modular nome used later. Set

$$u(q)=-\frac{2q(1+6q)}{1+8q},\qquad v(q)=\frac{16q^4}{1+8q},\qquad B(q)=\mathcal{H}(u(q),v(q)). \tag{4}$$

The target point corresponds to $q_0=1/12$.

Lemma 2. *On the curve (4),*

$$4q(2q+1)B+(8q^2+12q+1)U+(24q^2+8q-1)V=0. \tag{5}$$

Moreover, $L_B B=0$, where $\vartheta=\vartheta_q$ and coefficients in the following operator are placed to the left of powers of ϑ:

$$\begin{aligned}L_B=\ &-\vartheta^3-2q(22\vartheta^3+15\vartheta^2+9\vartheta+2)\\ &-4q^2(181\vartheta^3+219\vartheta^2+142\vartheta+32)\\ &-32q^3(169\vartheta^3+270\vartheta^2+181\vartheta+45)\\ &-64q^4(244\vartheta^3+384\vartheta^2+149\vartheta-18)\\ &+512q^5(\vartheta+1)(20\vartheta^2+148\vartheta+159)+49152q^6(\vartheta+1)(\vartheta+2)(2\vartheta+3).\end{aligned} \tag{6}$$

The analytic solution of this equation with constant term 1 is unique.

证明. We describe an exact certificate for both assertions. The coefficient ratios of (2) give $P_1\mathcal{H}=P_3\mathcal{H}=0$, with

$$\begin{aligned}P_1&=u(1-4u)\partial_u^2+v(1-16u)\partial_u\partial_v-12v^2\partial_v^2\\ &\quad+(1-10u)\partial_u-22v\partial_v-2,\\ P_3&=u\vartheta_v^3-v\vartheta_u(\vartheta_u+3\vartheta_v+1)(\vartheta_u+3\vartheta_v+2).\end{aligned} \tag{7}$$

For example, if $a_{j,k}$ is the coefficient in (2), then

$$\frac{a_{j+1,k}}{a_{j,k}}=\frac{2(2j+2k+1)(j+3k+1)}{(j+k+1)(j+1)},\qquad \frac{a_{j-1,k}}{a_{j,k-1}}=\frac{j(j+3k-2)(j+3k-1)}{k^3}.$$

The second equality is used for $j,k\ge 1$; the multipliers in P_3 make the boundary terms vanish.

A further, fourth-order radial equation is obtained as follows. Define

$$a_n(c)=\sum_{k=0}^{n}\frac{(n+2k)!}{(n-k)!(k!)^3}c^{n-k},\qquad F_c(z)=\sum_{n\ge 0}\binom{2n}{n}a_n(c)z^n=\mathcal{H}(cz,z).$$

Summing first over $j = n - k$ by the binomial theorem gives

$$A_c(t) := \sum_{n\ge 0} a_n(c)t^n = \frac{1}{1-ct}\,{}_2F_1\left(\frac{1}{3},\frac{2}{3};1;\frac{27t}{(1-ct)^3}\right). \tag{8}$$

The pulled-back Gauss equation is $\sum_{r=0}^{4} t^r P_r(\vartheta_t)A_c = 0$, where

$$\begin{aligned}
P_0(h) &= -h^2,\\
P_1(h) &= (c+27)h^2 + (5c+27)h + c + 6,\\
P_2(h) &= 3c\big((c+18)h^2 - 2ch - c - 2\big),\\
P_3(h) &= -c^3h(5h+3),\\
P_4(h) &= 2c^4(h+1)^2.
\end{aligned}$$

Here the temporary notation $P_r(h)$ denotes coefficient polynomials, not the partial differential operators in (7). Multiplying coefficients by $\binom{2n}{n}$ gives the fifth-order annihilator $M = \sum z^r M_r(\vartheta_z)$, where

$$M_r(h) = 4^r P_r(h)\frac{(h+\frac{1}{2})_r}{(h+1)_r}(h+r)(h+r-1)(h+r-2).$$

Each $M_r(h)$ simplifies to a polynomial. There is an exact Ore factorization $M = QL$, with

$$Q = \frac{-(1+8cz)\vartheta_z + 2(1+2cz)}{1-10cz}, \qquad L = \sum_{r=0}^{4} z^r \ell_r(\vartheta_z), \tag{9}$$

where

$$\begin{aligned}
\ell_0(h) &= h^3(h-1),\\
\ell_1(h) &= -2h\big((11c+54)h^3 + (-c+81)h^2 + (7c+39)h + c + 6\big),\\
\ell_2(h) &= 12c(2h+1)\big((7c+45)h^3 + 7ch^2 + (6c-5)h + 2c\big),\\
\ell_3(h) &= -8c^3(2h+1)(2h+3)(17h^2+24h+12),\\
\ell_4(h) &= 80c^4(h+1)(2h+1)(2h+3)(2h+5).
\end{aligned}$$

The rule $\vartheta_z f = f\vartheta_z + zf'$ verifies (9) coefficient by coefficient. Thus $E = LF_c$ has the form $E = Cz^2(1+8cz)^{-3/2}$. The first three coefficients of F_c are $1, 2c+12, 6c^2+144c+540$; substitution shows $[z^2]E = 0$. Consequently $LF_c = 0$.

Replace c by u/v, z by v, and ϑ_z by $\vartheta_u + \vartheta_v$ in this radial equation. Together with (7), it supplies the three equations used by the jet certificate. More precisely, the certificate gives rational functions of q expressing $L_B B$ as a linear combination of derivatives of these three equations of total order at most five. A second multiplier vector proves, for the left side W of (5),

$$\vartheta_q W = \frac{4q}{1+8q}W. \tag{10}$$

Both statements are checked by matching all 21 ordinary jets $\partial_u^a\partial_v^b\mathcal{H}$, $a+b\le 5$, over $\mathbb{Q}(q)$. The vectors are given in `prize_congruence_curve90_jet_certificate.json`; the independent verifier reconstructs the operators and checks every coefficient. Equation (10) gives $W = C\sqrt{1+8q}$, while $W(0) = 0$. This proves (5). Finally, $L_B|_{q=0} = -\vartheta_q^3$, so the coefficient of q^n, $n \ge 1$, is determined uniquely after division by $-n^3$. □

At q_0, (5) and the chain rule give

$$7B + 37U - 3V = 0, \qquad \vartheta_q B = \frac{14}{15}U + \frac{18}{5}V.$$

Eliminating U, V therefore reduces (3) to

$$S = 3\big[(35\vartheta_q + 18)B\big](1/12). \tag{11}$$

3 The level-60 modular transformation

For τ in the upper half-plane, write $\eta_d = \eta(d\tau)$ and $Q_\tau = e^{2\pi i\tau}$. Put $D = (2\pi i)^{-1}\mathrm{d}/\mathrm{d}\tau = Q_\tau \mathrm{d}/\mathrm{d}Q_\tau$. Define

$$\begin{aligned} \mathcal{A} &= \eta_1\eta_3\eta_4\eta_5\eta_{12}\eta_{15}\eta_{20}\eta_{60}, & \mathcal{B} &= \eta_2^2\eta_6^2\eta_{10}^2\eta_{30}^2, \\ X &= \mathcal{A}/\mathcal{B}, & Z &= \eta_2\eta_6\eta_{10}\eta_{30}/X^2. \end{aligned} \tag{12}$$

Write $\mathcal{C} = \eta_2\eta_6\eta_{10}\eta_{30}$ and $L = D\log X$. The product expansions give $X = Q_\tau + O(Q_\tau^2)$, $Z = 1 + O(Q_\tau)$, and $L = 1 + O(Q_\tau)$. The candidate polynomials are

$$\begin{aligned} G(x) &= (1-x)(1-4x)(1+4x^2)(1-x+4x^2), \\ J(x) &= 4x(1 - 8x + 40x^2 - 83x^3 + 160x^4 - 144x^5). \end{aligned} \tag{13}$$

3.1 An independent eta-product and Sturm proof

Proposition 3. *The eta quotients in (12) satisfy the identities*

$$L^2 = G(X)Z^2, \qquad 2ZD^2Z - 3(DZ)^2 = Z^4 J(X). \tag{14}$$

In particular, near the cusp $Q_\tau = 0$,

$$DX = X\sqrt{G(X)}Z, \qquad \sqrt{G(0)} = 1. \tag{15}$$

As a function of the local coordinate $x = X$, the normalized period $Z(x) = 1 + O(x)$ satisfies

$$L_Z Z = 0, \qquad L_Z = G\delta^3 + \frac{3}{2}(\delta G)\delta^2 + \left(\frac{1}{2}\delta^2 G - J\right)\delta - \frac{1}{2}\delta J, \quad \delta = x\frac{\mathrm{d}}{\mathrm{d}x}. \tag{16}$$

The analytic solution of (16) with constant term 1 is unique.

证明. We give all modularity and holomorphy hypotheses needed for the exact Sturm checks. For an eta quotient $f = \prod_{d|60}\eta_d^{r_d}$ of integer weight $k = \frac{1}{2}\sum r_d$, the classical eta-quotient criterion [4, 5] requires

$$\sum_{d|60} d r_d \equiv \sum_{d|60}(60/d)r_d \equiv 0 \pmod{24}. \tag{17}$$

Its character is the Kronecker character determined by $(-1)^k\prod d^{r_d}$. Thus, for even k, a rational square $\prod d^{r_d}$ gives trivial character. For $\mathcal{A}, \mathcal{B}, \mathcal{C}, X, Z$, respectively, the weights are $4, 4, 2, 0, 2$, both sums in (17) give the list $120, 96, 48, 24, 0$, and the corresponding products are

$12960000, 12960000, 3600, 1, 3600$. These are all squares, and the two congruences hold for each quotient. Eta has no zeros in the upper half-plane, so X is a modular unit there.

Put

$$S_+ = \{1, 3, 4, 5, 12, 15, 20, 60\}, \qquad S_- = \{2, 6, 10, 30\}.$$

The exponents of X are 1 on S_+ and -2 on S_-; those of Z are -2 and 5, respectively. The cusp-order formula [5], for a cusp of denominator $c \mid 60$, is

$$v_c(f) = \frac{60}{24\gcd(c^2, 60)} \sum_{d|60} \frac{\gcd(c, d)^2}{d} r_d. \tag{18}$$

Here the order is with respect to the local parameter of that cusp. There is one cusp for each divisor c: the multiplicity $\varphi(\gcd(c, 60/c))$ is 1 in every case. Formula (18) gives the following complete table, grouped according to the twelve cusp denominators:

denominators c	$v_c(X)$	$v_c(Z)$	$v_c(G(X)Z^2)$	$v_c(J(X)Z^4)$
$2, 6, 10, 30$	-2	6	0	12
$1, 3, 4, 5, 12, 15, 20, 60$	1	0	0	1

For example, at a pole of X, both G and J have degree 6, so the last two orders are $6(-2)+2(6) = 0$ and $6(-2)+4(6) = 12$. At a zero of X, use $G(0) = 1$ and the simple zero of J at 0. Consequently Z is a holomorphic weight-2 form, and $G(X)Z^2$ and $J(X)Z^4$ are holomorphic forms of weights 4 and 8 on $\Gamma_0(60)$ with trivial character.

The logarithmic derivative of the eta product gives

$$L = \frac{1}{24} \sum_{d|60} r_d d E_2(d\tau), \qquad E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) Q_\tau^n, \tag{19}$$

where r_d is the exponent vector of X and $\sigma_1(n) = \sum_{a|n} a$. The E_2 transformation anomaly cancels because $\sum r_d = 0$, so L has weight 2. Alternatively this follows directly by differentiating the weight-0 transformation of X. It is holomorphic in the upper half-plane. For any cusp represented by $\sigma \in \mathrm{SL}_2(\mathbb{Z})$, write its width as w, and $X(\sigma\tau) = q_c^m(a_0 + O(q_c))$, where $q_c = e^{2\pi i\tau/w}$ and $a_0 \neq 0$. Then

$$(L|_2\sigma)(\tau) = D \log X(\sigma\tau) = \frac{m}{w} + O(q_c).$$

This proves holomorphy at all cusps, including poles of X. Hence $L^2 - G(X)Z^2$ is a holomorphic weight-4 form.

For completeness, the derivative expression $R(f) = 2fD^2f - 3(Df)^2$ of a holomorphic weight-2 form is holomorphic of weight 8. It equals one third of the second Rankin–Cohen bracket $[f, f]_2$. Directly, for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $j = c\tau + d$, and $\alpha = c/(2\pi i)$, differentiation gives

$$\begin{aligned} f(\gamma\tau) &= j^2 f(\tau), \\ (Df)(\gamma\tau) &= j^4 Df(\tau) + 2\alpha j^3 f(\tau), \\ (D^2 f)(\gamma\tau) &= j^6 D^2 f(\tau) + 6\alpha j^5 Df(\tau) + 6\alpha^2 j^4 f(\tau). \end{aligned}$$

Substitution cancels the anomaly terms and gives $R(f)(\gamma\tau) = j^8 R(f)(\tau)$. The same calculation for the slash operator at any cusp gives $R(f)|_8\sigma = R(f|_2\sigma)$. The right side has a holomorphic

local Fourier series. This proves the cusp holomorphy as well as covariance. Apply this to $f = Z$. Thus $R(Z) - Z^4 J(X)$ is a holomorphic weight-8 form.

The index is

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(60)] = 60(1 + 1/2)(1 + 1/3)(1 + 1/5) = 144.$$

The Sturm bounds [6] for the two differences are therefore $4 \cdot 144/12 = 48$ and $8 \cdot 144/12 = 96$. The self-contained program `sun90_modparam_verify.py` proves that their exact integer coefficients vanish from degree 0 through 48 and 96, respectively. It also computes (17), every cusp order, and the derivative anomaly cancellation independently from the displayed exponent vectors.

Here is a reproducible specification of the integer-series arithmetic. For $f = Q_\tau^m \sum_{n\geq 0} u_n Q_\tau^n$, $u_0 = 1$, the eta product gives

$$b_n = -\sum_{\substack{d|60 \\ d|n}} r_d d\, \sigma_1(n/d), \qquad n u_n = \sum_{k=1}^{n} b_k u_{n-k}. \tag{20}$$

Each division is checked exact. Ordinary integer convolutions give the products, and D multiplies the coefficient of Q_τ^n by n. The polynomial coefficients used in the checks are

$$G = (1, -6, 17, -48, 68, -96, 64),$$
$$J = (0, 4, -32, 160, -332, 640, -576),$$

listed in ascending powers. Sturm's theorem now proves both complete identities in (14). Equivalently one can apply its congruence formulation for every prime to these integral coefficients. Thus these checks are finite certificates for identities, with their modularity hypotheses supplied above.

Near the cusp, L, Z, and the square root of G all have constant term 1, proving (15). Since $X = Q_\tau + O(Q_\tau^2)$, we may use $x = X$ as a local coordinate. Applying the chain rule to (14) gives

$$2GZ\delta^2 Z - G(\delta Z)^2 + (\delta G) Z \delta Z - J Z^2 = 0.$$

Its δ derivative is

$$Z\big[2G\delta^3 Z + 3(\delta G)\delta^2 Z + (\delta^2 G - 2J)\delta Z - (\delta J) Z\big] = 0.$$

Because $Z(0) = 1$, division by $2Z$ proves (16). Finally its constant-coefficient indicial operator is δ^3: the coefficient of x^n is determined recursively by division by n^3 for every $n \geq 1$. This proves uniqueness. □

3.2 The algebraic pullback

Set

$$\phi(x) = \frac{x}{2(1-2x)^2}, \qquad K(x) = (1-2x)^2\sqrt{1+4x^2}. \tag{21}$$

An exact differential-operator identity is

$$K^{-1} L_B(\phi(x), \vartheta_q) K = \frac{(1+4x^2)^2}{(2x-1)^9(2x+1)} L_Z, \qquad \vartheta_q = \frac{x(1-2x)}{1+2x}\frac{\mathrm{d}}{\mathrm{d}x}. \tag{22}$$

To verify it, use $K'/K = -4/(1-2x)+4x/(1+4x^2)$ and compare the four coefficients of $(\mathrm{d}/\mathrm{d}x)^r$, $0 \le r \le 3$. These are rational-function identities, all checked in the attached verifier. Since $\phi(0) = 0$, $\phi'(0) = 1/2$, and $K(0) = 1$, uniqueness in Lemma 2 yields

$$B(\phi(x)) = K(x)Z(x). \tag{23}$$

4 An independently certified modular equation and CM point

We next establish the needed special value of X, rather than assuming its entry in a table. Define the symmetric polynomial

$$\Phi(x,y) = x^8 + y^8 + \sum_{i=1}^{7}\sum_{j=1}^{i} c_{ij}S_{ij}(x,y), \tag{24}$$

where $S_{ii} = x^i y^i$ and $S_{ij} = x^i y^j + x^j y^i$ for $i > j$. Its complete coefficient array is

$i\backslash j$	1	2	3	4	5	6	7
1	−1						
2	7	−14					
3	−21	49	−392				
4	49	−119	588	−168			
5	−77	231	−1120	2352	−6272		
6	77	−245	924	−1904	3136	−3584	
7	−56	308	−1232	3136	−5376	7168	−4096

Lemma 4. *The identity $\Phi(X(\tau), X(7\tau)) = 0$ holds. At $\tau_0 = i\sqrt{7/60}$,*

$$X(\tau_0) = x_0 := \frac{5-\sqrt{21}}{4}. \tag{25}$$

The branch $y = y(x)$ of $\Phi(x,y) = 0$ through (x_0, x_0) satisfies

$$y'(x_0) = -1, \qquad y''(x_0) = \frac{56}{15} + \frac{32\sqrt{21}}{63}. \tag{26}$$

证明. The eta-quotient criterion shows that $\mathcal{A}$ and $\mathcal{B}$ are holomorphic weight-4 forms of trivial character on $\Gamma_0(60)$. For their exponent vectors, respectively, both sums $\sum dr_d$ and $\sum(60/d)r_d$ equal 120 and 96; their products $\prod d^{r_d}$ are squares. All exponents are nonnegative. Thus

$$\mathcal{B}(\tau)^8\mathcal{B}(7\tau)^8\Phi(X(\tau), X(7\tau))$$

is a holomorphic weight-64 form of trivial character on $\Gamma_0(420)$. The index of this group is 1152, so its Sturm bound is $64 \cdot 1152/12 = 6144$. The exact integer eta-product calculation in `prize90_geometry_mod7_sturm.py` proves that every coefficient from Q_τ^0 through Q_τ^{6144} is zero. Sturm's theorem [6] proves the modular identity. The computation uses exact integer convolution with a rigorous coefficient bound; its result is not a finite numerical test substituted for a modular identity.

The eta transformation $\eta(-1/\tau) = \sqrt{-i\tau}\eta(\tau)$, together with the reciprocal symmetry of the exponents of X, gives

$$X\left(-\frac{1}{60\tau}\right) = X(\tau). \tag{27}$$

Put $\tau_1 = i/\sqrt{420}$. Then $7\tau_1 = -1/(60\tau_1) = \tau_0$, so $X(\tau_0)$ lies on the diagonal of the modular equation. Direct factorization gives

$$\begin{aligned}\Phi(x,x) = & -x^2(2x-1)^2(4x^2-10x+1)(4x^2+2x+1)\\ & \cdot(4x^3+2x-1)(16x^3-8x^2-1).\end{aligned} \tag{28}$$

For completeness, the correct root is selected by rational inequalities. Machin's formula with alternating-series bounds, followed by positive Taylor bounds for the exponential, gives

$$\frac{29}{250} < e^{-2\pi\sqrt{7/60}} < \frac{59}{500}.$$

One may use $6283/2000 < \pi < 22/7$ and $683/2000 < \sqrt{7/60} < 427/1250$, and twenty exponential Taylor terms. Let $l = 29/250$, $h = 59/500$, and put

$$S_A = \sum_{d\in\{1,3,4,5,12,15,20,60\}} \frac{h^d}{1-h^d}, \qquad S_B = 2\sum_{d\in\{2,6,10,30\}} \frac{h^d}{1-h^d}.$$

Using $\prod(1-a_n) \geq 1 - \sum a_n$ first for finite products and then by a limit, the eta products imply

$$l(1-S_A) < X(\tau_0) < \frac{h(1-l)}{1-S_B}.$$

Exact rational comparison makes the lower bound greater than $1/10$ and the upper bound smaller than $11/100$. Only one root of (28) lies in this interval: it is the smaller root of $4x^2-10x+1$. This proves (25). These inequalities and the root count are independently checked by `prize90_geometry_cm_verify.py`.

Finally $\Phi_x(x_0,x_0) \neq 0$. Symmetry gives $y'(x_0) = -1$, and implicit differentiation gives

$$y''(x_0) = -\frac{2(\Phi_{xx}-\Phi_{xy})}{\Phi_x}(x_0,x_0).$$

Substitution of the explicit polynomial (24) proves (26) in the quadratic field $\mathbb{Q}(\sqrt{21})$. □

5 The weighted period value by Fricke differentiation

Here we derive the weighted evaluation directly, so that no specific CM table or displayed Ramanujan–Sato formula is needed. Write

$$D = \frac{1}{2\pi i}\frac{\mathrm{d}}{\mathrm{d}\tau}, \qquad g(x) = \sqrt{G(x)}.$$

Proposition 3 gives

$$DX = Xg(X)Z \tag{29}$$

on the branch from the cusp. We check its sign at the required CM point without a table convention. Lemma 4 proves $0 < Q_0 = e^{-2\pi\sqrt{7/60}} < h = 59/500$. All eta factors, and hence Z, are positive for real $0 < Q_\tau < 1$. The elementary inequality

$$\sum_{n\geq 1} \sigma_1(n) t^n \leq \frac{t}{(1-t)^3} \quad (0 < t < 1)$$

follows from $\sigma_1(n) \leq n(n+1)/2$. Dropping the positive S_- terms in (19) gives, for $0 < Q_\tau \leq h$,

$$L(Q_\tau) \geq 1 - \sum_{d\in S_+} \frac{dh^d}{(1-h^d)^3} > 0. \tag{30}$$

The last comparison is exact rational arithmetic, also checked by `sun90_modparam_verify.py`. Thus $L = +\sqrt{G(X)}Z$ at τ_0, with $g(x_0) > 0$. This also shows that X is strictly increasing on the real nome interval from the cusp to Q_0, so that the local coordinate and period in Proposition 3 reach this CM value on the chosen branch. The eta transformation and (27) give

$$Z\left(-\frac{1}{60\tau}\right) = -60\tau^2 Z(\tau). \tag{31}$$

Set $t = \tau_1$, $s = 7t = \tau_0$, and abbreviate $Z_s = Z(s)$. Differentiating (27) and (31) yields

$$Z(t) = 7Z_s, \qquad DX(t) = -7DX(s), \qquad DZ(t) + 49DZ(s) = \frac{14\sqrt{105}}{\pi} Z_s. \tag{32}$$

In particular the two occurrences of DX have opposite signs; at t one must use $DX(t) = -x_0 g(x_0) Z(t)$.

The branch $y(x) = X(7\tau)$, $x = X(\tau)$, has

$$y''(x_0) = \frac{49D^2X(s) + D^2X(t)}{DX(t)^2}.$$

At s and t, respectively, differentiating (29) with the appropriate signs gives

$$D^2X(s) = x_0 g(g + x_0 g') Z_s^2 + x_0 g\, DZ(s),$$
$$D^2X(t) = x_0 g(g + x_0 g') Z(t)^2 - x_0 g\, DZ(t),$$

where g, g' are evaluated at x_0. Substitute (32), and use $DZ(s) = x_0 g Z_s Z_x(x_0)$. The result is

$$y''(x_0) = \frac{2}{x_0} + \frac{G'(x_0)}{G(x_0)} + \frac{2Z_x(x_0)}{Z_s} - \frac{14\sqrt{105}}{49\pi x_0 g(x_0) Z_s}. \tag{33}$$

Consequently, if

$$\lambda = 1 + \frac{x_0 G'(x_0)}{2G(x_0)} - \frac{x_0 y''(x_0)}{2} = \frac{3}{2} - \frac{11\sqrt{21}}{42}, \tag{34}$$

then (33) proves the exact evaluation

$$[(\delta + \lambda)Z](x_0) = \frac{\sqrt{60/7}}{2\pi\sqrt{G(x_0)}}. \tag{35}$$

This derivation uses the independent modular equation and its implicit derivatives from Lemma 4.

6 Completion of the proof and convergence

Since $\phi(x_0) = 1/12$, differentiate (23) to obtain

$$(35\vartheta_q + 18)B = \frac{35K(x)(1-2x)}{1+2x} \cdot \left[\delta + \delta \log K(x) + \frac{18(1+2x)}{35(1-2x)}\right] Z(x). \tag{36}$$

At $x = x_0$, the constant inside the square brackets equals λ in (34). All these equalities are exact in $\mathbb{Q}(\sqrt{21})$. The positive algebraic factor

$$\frac{35K(x_0)(1-2x_0)}{1+2x_0} \frac{\sqrt{60/7}}{2\sqrt{G(x_0)}}$$

has square 900. Since $0 < x_0 < 1/4$, every displayed square root and remaining factor is positive, so this factor equals 30. Equations (35) and (36) now give $[(35\vartheta_q+18)B](1/12) = 30/\pi$. Combined with (11), this proves (1).

We justify all analytic operations and continuation used above. Replacing -324 by 324 majorizes the original double series termwise. From $\binom{2n}{n} \le 4^n$ and (8), at $z_0 = 1/2160$ the corresponding Gauss parameters are

$$1296z_0 = \frac{3}{5} < 1, \qquad \frac{108z_0}{(1-1296z_0)^3} = \frac{25}{32} < 1.$$

The strict inequalities hold in a neighborhood, proving local uniform absolute convergence, including differentiated series. A fully rational majorant is available at $R = 1/2088 > z_0$:

$$1 - 1296R = \frac{11}{29}, \qquad \frac{108R}{(1-1296R)^3} = \frac{2523}{2662}, \qquad \sum_n \left|\binom{2n}{n} a_n(-324)\right| R^n \le \frac{7018}{139}.$$

For example, the absolute tail of (1) after $n = N-1$ is at most

$$\frac{7018}{139}\left(\frac{29}{30}\right)^N \big(30(357N+103)+310590\big).$$

Along $0 \le q \le 1/12$, both $|u(q)|$ and $v(q)$ increase, so this entire path lies in the same domain of the bivariate series. The leading coefficient of (6) in ϑ_q^3 is

$$(2q-1)(6q+1)(8q+1)^3(16q+1),$$

which has no zero on $0 < q \le 1/12$. For $0 \le x \le x_0$, $K(x)$ and $\sqrt{G(x)}$ take their positive branches, and no singularity of (16) is crossed. The identities proved near the origin therefore extend to the required point.

7 Exact certificates and disclosure

From the repository root, run the complete verification entry point:

```
python ramanujan-series/sun90-identity/verify.py
```

It stages the programs in `code/` and the inputs in `certificates/` into a temporary flat directory, runs all four components, and leaves tracked files unchanged. Python 3.9 or later and SymPy 1.14.0 are required. Component JSON reports named below are generated in that temporary directory; their contents are retained under `component_results` in `reproducibility/full_verification.json`. The ancillary files have distinct roles. The differential-algebra verification is performed by `verify_sun90.py`, using only `prize_congruence_curve90_jet_certificate.json` alongside it in the staged directory. This component checks the coefficient-ratio PDEs, the Gauss pullback and Ore factorization, all jet multipliers, the algebraic modular pullback, and the final constants and convergence majorant. Its modular special-value comparison is consistent with (35). The general modular data and the special value are established independently of that comparison by the eta/Sturm and Fricke proofs above, with the following additional certificates:

- `sun90_modparam_verify.py` independently computes the eta modularity congruences, characters and all twelve cusp orders, checks the weight-4 and weight-8 identities through the complete Sturm bounds 48 and 96, and verifies the derivative covariance, the linear period equation, and the positive sign bound (30). It imports no earlier project script.
- `sun90_modparam_result.json` records these exact results, the cusp table, and the Fourier-series hashes.
- `sun90_modparam_proof.txt` gives a separate prose account of the modularity hypotheses and the same independent proof.
- `prize90_geometry_mod7_result.json` specifies the polynomial (24). The discovery script alone is not its proof.
- `prize90_geometry_mod7_sturm.py` proves the modular identity through the complete Sturm bound; its result is recorded in `prize90_geometry_mod7_sturm_result.json`.
- `prize90_geometry_cm_verify.py` checks the rational nome bounds, isolates the CM root, and verifies the implicit derivatives and (34); its result is recorded in `prize90_geometry_cm_verify_result.json`.
- `verify_sun90_full.py` runs the independent general parametrization, algebraic bridge, degree-seven Sturm, and CM programs in sequence, checks that their conclusions agree, and writes `sun90_full_verification.json`.

The independent general-parametrization component can also be run alone from the staged directory with `pythonsun90_modparam_verify.py`. The separate degree-seven Sturm certificate checks coefficients through degree 6144, including the constant term, at weight 64 and level 420. Its polynomial string has SHA-256 digest

```
b24469bdc909d58bce0e29b1dc0e51b26
a9c47d98d5b19a7f8f6d5cad43fa768.
```

The table in (24) also specifies this polynomial without relying on a digest. No floating-point root recognition is used in the proof.

AI assistance and submission status. OpenAI Codex assisted with symbolic exploration, organization of the argument, and preparation of the exact verification programs and this manuscript. The author is 李想. The external mathematical inputs and the executable certificates are disclosed above. The discovery computations, including exploratory geometric and numerical calculations, are not assumptions of the proof. Version 1 was emailed to the problem's author on 16 October 2026. This public Version 2 has not been emailed separately. Independent external review, first-proof priority, and prize eligibility remain unconfirmed.

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